

Shail K Patel

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github.com/shailkpatel

shailkpatel.github.io

Summary

AI Engineer with 1.5 years in production ML, currently the founding engineer on an LLM product. Designed and shipped its RAG, document extraction, and agent systems, cutting per-invoice inference cost 75%.

Experience

Founding Machine Learning Engineer

RestaurantPilot.ai (Restaurant Tech)

Seattle, USA (Remote)

Nov. 2025 - present

- Built production RAG with hybrid vector and keyword retrieval and fused reranking over ChromaDB and MongoDB Atlas Vector Search.
- Own 5 LLM extraction pipelines pairing OCR and PDF parsing with schema-constrained structured output; cut per-invoice inference cost 75% and LLM call latency 59%.
- Built a tool-calling agent loop over 38 internal tools and a provider-agnostic layer routing across multiple LLM providers; lead a 4-person team across ML, backend, and frontend.

Machine Learning Engineer

GET MY SPACE (Parking Management Startup)

Ahmedabad, India

Jan. 2025 - Jun. 2025

- Built a real-time ANPR pipeline over live camera feeds, with format-validated fallback across multiple recognition models, improving accuracy 73%.
- Designed the system's database schema and deployed computer vision models behind production APIs, cutting recognition latency 34%.

Projects

Open to Work - LLM Job Matching and Resume Tailoring

Oct 2026

github.com/shailkpatel

- Built a FastAPI app that ingests GitHub repositories and resumes, uses LLMs to extract a per-project skill profile, and parses job postings from pasted text or screenshots.
- Scores each posting against existing resumes or generates a tailored one, with skill-gap analysis and a pay-scale view across target roles.

PredictGrad - Academic Risk Detection with ML

Jun 2025

predictgrad.streamlit.app

- Developed a two-stage, leakage-aware pipeline using prior-semester performance to forecast Semester 3 marks, then classify students at risk of significant academic decline before the semester begins.
- Evaluated 274 model pipelines with Bayesian optimization and ensemble methods, achieving a best test MAE of 5.71 on a 0 - 100 scale and 0.657 recall for risk detection under class imbalance.

This project led to a published preprint: A Two-Stage, Leakage-Aware Framework for Early Academic Risk Detection in Undergraduate Engineering Cohorts (doi.org/10.5281/zenodo.19686376).

Education

LJ University, Ahmedabad

B.Eng in Artificial Intelligence and Machine Learning

Expected Jun. 2027

Skills

LLM & GenAI: LLMs, RAG, Generative AI, Agentic AI, prompt engineering, structured extraction, evaluation datasets, LangChain, LangGraph, LlamaIndex, LiteLLM, Hugging Face, OpenAI & Gemini APIs

Retrieval: ChromaDB, MongoDB Atlas, embeddings, hybrid retrieval, reranking

Backend & Infra: Python, FastAPI, Django, REST APIs, Docker, AWS (S3, Lambda), Git, Linux

Data & ML: PostgreSQL, MongoDB, Redis, SQL, NOSQL, Pandas, NumPy, PyTorch, Scikit-learn, NLP, Computer Vision, MLOps, model deployment